INDIA.RUNS

Build what next India runs on

Team Name : teamnextmatrix

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Problem Statement : Intelligent Candidate Discovery & Ranking (Offline-First CPU Streaming Engine)

Solution Overview

Proposed Solution

- **Offline-First CPU Streaming Engine:** A lightweight, high-performance Python system designed to process 100K candidates without GPU or network APIs.
- **Zero-Tolerance Anomaly Guard:** Identifies and excludes trap candidates directly at the ingestion layer.
- **Platform Signal Modifier:** Integrates behavioral metrics (responsiveness, activity) directly into relevance weights.

Key Differentiators

- **High Latency-Quality Tradeoff:** Full pool processed in <15 seconds (limit: 5 minutes) using clean generators.
- **Hallucination Prevention:** Factual dynamic reasoning directly compiled from candidate profile parameters.
- **Anti-Keyword-Stuffing:** Normalizes skill proficiency and endorsements against verified career histories.

JD Understanding & Candidate Evaluation

Key JD Requirements

- **Experience:** Target 5–9 years of applied ML/AI experience at product companies.
- **Core Tech Depth:** Dense retrieval, vector databases, Python, and search evaluation (NDCG, MAP).
- **Vibe & Vocation:** Product-shipper mindset, relocation to Noida/Pune, short notice period.

Evaluation Beyond Keywords

- **Career Audit:** Excludes candidates spending their entire career in consulting/IT-services firms.
- **Expertise Verification:** Validates listed skills using duration months and endorsements.
- **Signal Envelope:** Weights actual responsiveness (rate $\geq 70\%$) and recent activity (last active ≤ 90 days).

Ranking Methodology

Scoring Components

- **Base Score (Max 80 pts):** Core skills (40 pts) + Nice-to-haves (20 pts) + Experience Fit (20 pts).
- **Disqualifiers (Multipliers):** Applies fractions (e.g. 0.1x) for consulting-only background, CV-only skills, or pure research.
- **Behavioral Modifiers:** Multiplies by response rate, notice period, active status, and GitHub score.

Tie-Breaking and Sorting

- **Rounding Match:** Scores rounded to 4 decimal places to match the CSV format exactly.
- **Deterministic Tie-Breaker:** Ties broken using candidate_id ascending (e.g., CAND_0001 before CAND_0002) as required by the spec.

Explainability & Data Validation

Explainability & Non-Hallucination

- **Dynamic Fact Reasoning:** Combines actual profile statistics (YoE, title, matched skills, notice period) into a structured sentence.
- **No Templates:** Varied sentences based on candidate conditions to prevent flat-review flags.

Honeypot Filter Rules

- **Skill Anomalies:** Flags and rejects "expert" proficiency skills listed with 0 months.
- **YoE Mismatches:** Rejects profiles with YoE discrepancies (difference >3 years from career history sum).
- **Tenure Checks:** Catches impossible timelines (e.g., claiming 8 years at Krutrim/Sarvam AI).

End-to-End Workflow

The system runs linearly to maintain constant memory complexity O(1):

Input JSONL Streaming → Honeypot Guard (Skip Traps) → Feature & Match Scoring → Heuristic Multipliers → Deterministic Tie-Break Sort → Output CSV & Reasoning

System Architecture

Ingestion & Filtering Layer

- Streams JSON lines sequentially, parsing profile, career_history, skills, and redrob_signals.
- Applies boolean validation guards to weed out 151 honeypots.

Scoring & Compiler Layer

- Computes baseline fit, evaluates qualifiers/location matching, and factors in behavioral multipliers.
- Ties are resolved dynamically, and top 100 elements are written with structured facts into a CSV writer.

Results & Performance

Compute Constraints

- **Execution Time:** 15 seconds (Limit: 5 minutes) for 100K lines.
- **Memory Limit:** <50MB RAM (Limit: 16GB).
- **Offline Check:** 100% offline, zero network requests, CPU-only.

Top Ranked Results

Rank	ID	Score	Title	Company	Location
1	CAND_0052328	1.0000	Recommendation Eng	Amazon	Pune
2	CAND_0018499	0.9990	Senior ML Engineer	Zomato	Noida
3	CAND_0068351	0.9977	Lead AI Engineer	Sarvam AI	Delhi

Technologies Used

Python Standard Library (Pure & Dependency-Free)

To guarantee maximum execution speed, zero compatibility regression, and lightweight container deployment, we utilized Python's standard library modules:

- **json:** For fast streaming line-by-line parsing of 100K JSONL records.
- **csv & argparse:** For robust command-line handling and strict schema-compliant CSV writing.
- **datetime:** For parsing relative candidate last active dates and tenure validation.
- **PyYAML:** Used exclusively for metadata parsing in repository configs.

Submission Assets

GitHub Code Repository

Our code repository contains setup files, clean execution commands, and configuration specifications:

`https://github.com/sanoop6027/ai_recruiter_challenge`

Output Artifacts

- **Ranked Shortlist:** `teamnextmatrix.csv` (exactly 100 rows, score non-increasing, tie-broken).
- **Sandbox Link:** Google Colab notebook for sample execution verification.

THANK YOU

teamnextmatrix | Anoop Singh & Rohit Sharma